

Limits and Continuity II

1. For which of the following functions f is there a continuous function g with domain $\mathbb{R}$ such that $g(x) = f(x)$ for all x in the domain of f ?

i) $f(x) = \frac{x^2-9}{x+3}$

ii) $f(x) = x^2 \sin \frac{1}{x}$

iii) $f(x) = \frac{|x|}{x}$

2. Show that the polynomial

$$p(x) = x^3 + 2x + 1$$

has at least one root in the closed interval $[-1, 0]$.

3. For a function $f : D \rightarrow \mathbb{R}$, the point $p \in D$ that satisfies

$$f(p) = p$$

is called a *fixed point* of f . Show that a continuous function $f : [-1, 1] \rightarrow \mathbb{R}$ has a fixed point in the closed interval $[-1, 1]$ if $-1 \leq f(x) \leq 1$ for $x \in [-1, 1]$.

Hint: Consider $g(x) = f(x) - x$.

4. Using the definition of a derivative, find the derivatives of the following functions.

$$f(x) = x^3, \quad g(x) = \frac{1-x}{x+2}, \quad h(x) = \sqrt{x^2+1}$$

5. (a) Using the definition of a derivative, show that for any nonnegative integer $n \geq 0$,

$$\frac{d}{dx} x^n = nx^{n-1}.$$

(b) Let f and g be differentiable functions, and let $h(x) = af(x) + bg(x)$ for $a, b \in \mathbb{R}$. Show that

$$h'(x) = af'(x) + bg'(x).$$

(c) Using parts (a) and (b), conclude that all polynomials with real coefficients are differentiable on $\mathbb{R}$.

6. Show that not all continuous functions are differentiable by providing a counterexample.